



Effect of pulse consumption on perceived flatulence and gastrointestinal function in healthy males

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ABSTRACT

Despite their demonstrated health benefits, consumption of pulses in Western societies has traditionally been low. This is, in part, due to the perception that pulses cause flatulence and gastrointestinal upset. This randomized, double-blind placebo-controlled, cross-over study assessed the impact of 28 consecutive days consumption of 100 g dry weight Kabuli chickpeas, green Laird lentils, and green peas, in comparison to a potato control, on perceived flatulence, abdominal comfort, bowel movements and overall gastrointestinal function using a questionnaire with yes/no and visual analog rankings, in 21 healthy males between the ages of 19–40. Questionnaires were completed during three phases (PRE, EARLY, LATE) of each treatment period. Chi-square (occurrence) and Wilcoxon test (severity rank) analyses revealed minor changes in occurrence and/or severity of flatulence and abdominal comfort, but no changes in overall gastrointestinal function. These results demonstrate that pulses containing oligosaccharides are well tolerated with negligible perceived changes in flatulence and overall gastrointestinal function when incorporated into the diet of healthy adult males.

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1. Background

Pulses are the edible seed portion of leguminous plants, including chickpeas, lentils, green peas and a wide variety of beans (Pulse Canada, 2009). While these are staple foods, particularly in Latin America (Leterme & Carmenza Munoz, 2002) and India (Schneider, 2002), pulse consumption has traditionally been low in Western regions (Messina, 1999; Schneider, 2002), in large part, due to concerns that they cause flatulence (Minorsky, 2003) and, until relatively recently, pulses have seldom been recommended by health-promoting organizations (Leterme, 2002). There is, however, extensive evidence that pulses have numerous nutritional, health, and economic benefits. Pulses are low in fat, an inexpensive source of protein, rich in vitamins and minerals, and contribute to a high-fiber diet, which has been associated with a reduced risk of chronic disease, including diabetes and cardiovascular disease (Leterme, 2002; Slavin, 2008; Tharanathan & Mahadevamma, 2003). Pulses have also been investigated as functional foods (Hellendoorn, 1979; Pittaway, Ahuja, Robertson, & Ball, 2007; Rochfort & Panozzo, 2007; Tharanathan & Mahadevamma, 2003), particularly with respect to their potential as prebiotics (Guillon & Champ,

2002; Martinez-Villaluenga, Frias, & Vidal-Valverde, 2008) based on their constituent galacto-oligosaccharide sugars (GOS) (Macfarlane, Steed, & Macfarlane, 2008). Alpha-galactosides, especially those of the raffinose family, including stachyose and verbascose (Minorsky, 2003), are abundant in pulses, ranging from 2 to 10 g/100 g dry weight (Guillon & Champ, 2002). Paradoxically, these raffinose family GOS have also been identified as the primary flatulence-producing compounds in legumes (Granito, Champ, David, Bonnet, & Guerra, 2001; Granito, Michel, Frias, Champ, & Guerra, 2005; Martinez-Villaluenga et al., 2008).

In order to improve the acceptability of pulses and legumes, it has been common practice to remove the GOS (Garg, Banerjee, Verma, & Abraham, 1980; Ruperez, 1998) by overnight soaking followed by pressure cooking (discarding both soaking and cooking liquids), irradiation, germination, and enzyme treatments (Minorsky, 2003). These practices, once thought to improve the nutritional quality of pulses, must be reconsidered in light of the potential health benefits offered by GOS and pulses. Also, while there is evidence that pulses are gas-producing (Desrochers & Brauer, 2001; Granito et al., 2005; Leterme, 2002; Leterme & Carmenza Munoz, 2002; Minorsky, 2003; Phillips, 1993; Ruperez, 1998; Schneider, 2002), it is noteworthy that gas production is a normal physiologic process that, to some extent, aids in digestion by softening and helping to move stools through the colon (Hellendoorn,

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1979). Furthermore, there is evidence that a tolerance may be developed to excess flatulence produced, or GI symptoms associated with the consumption of beans (Fleming, O'Donnell, & Perma, 1985; O'Donnell & Fleming, 1984) and other low-digestible carbohydrates (LDC), including isolated GOS (Carabin & Flamm, 1999; Livesey, 2001; Marteau & Seksik, 2004). Similarly, according to the American Dietetic Association (Grabitske & Slavin, 2008) the laxative effect associated with the moderate consumption of LDCs, including fiber, is a desirable one, and GI disruptions such as flatulence, bloating and abdominal discomfort occur only when fiber is consumed in excess. Furthermore, there is evidence that, with regular consumption, the prebiotic action of pulses leads to alterations in gut microflora, which could, in turn, influence gastrointestinal symptoms (Roberfroid & Slavin, 2000; Swennen, Courtin, & Delcour, 2006).

To date, no research has focused directly on the level of tolerance to flatulence or other GI symptoms associated with the introduction of pulses into the Western diet. Furthermore, there may be differences in GI symptoms associated with different pulse varieties (Fleming, 1981). Some studies demonstrating an association between modest intakes of pulses and flatulence have noted that effects are mild (Bouhnik et al., 1999; O'Donnell & Fleming, 1984). In contrast, one study reported a 10-fold increase in flatulence when a diet consisting of 59% beans was compared to a habitual non-bean diet (Steggerda & Dimmick, 1966). Such a high level of inclusion of pulses in the diet is unlikely to be desirable or practical in the general population. More reasonably, brownies, in which up to 50% of the shortening was replaced with a cannellini bean puree, were shown to be nutritionally superior (i.e. lower in fat, higher in protein, fiber, vitamins and minerals) to a control brownie, with equal acceptability (Szafranski, Whittington, & Besinger, 2005), suggesting the potential for pulses to be used in the formulation of novel functional foods.

Further studies are needed to examine the possible GI effects of moderate pulse consumption, including those which examine a variety of pulse types as there may be differences in GI symptoms associated with different pulses (Fleming, 1981). Therefore, the purpose of this investigation was to determine the influence of regular consumption of a moderate amount of chickpeas, lentils, and green peas, on perceived GI function in a sample of healthy adult males. GI symptoms probed included those related to flatulence, abdominal comfort, bowel movements, and overall GI function. It was hypothesized that pulse consumption would be associated with GI disturbances which would lessen throughout the consumption period, in accordance with an adaptive response. It was also anticipated that differences could occur between the pulses, related to differences in fiber and the GOS constituents.

2. Methods

2.1. Study design

The study employed a randomized double-blind, placebo-controlled cross-over design consisting of four 28-day treatment periods separated by 28-day washout periods. Participants reported to the Human Nutraceutical Research Unit (HNRU) at the University of Guelph on Study Days –7, 1, 8, 15, 22 and 29 of each treatment period. The study was approved by the Human Research Ethics Board of the University of Guelph, and all participants provided written informed consent.

2.2. Participants

Healthy adult males were recruited from the University of Guelph campus and surrounding area. Participants were healthy,

non-smokers between the ages of 19 and 40 with a body mass index (BMI) 20–30 kg/m² and were not taking any medications. Exclusionary criteria were as follows: females, males <19 or >40 years of age, BMI <20 or >30.0 kg/m², allergy to pulses, pulse consumption >3 times per week, low fiber (<10 g per day) diets, history of GI disorders (including Celiac's or Crohn's disease, irritable bowel syndrome, acid reflux, ulcerative colitis, chronic constipation or diarrhea, and lactose intolerance), use of recreational drugs, use of prescription medications, use of antibiotics within 3 months prior to study commencement, use of anti-diarrheal agents or laxatives, use of natural health products (within 1 month prior to study commencement), and elite athletes. Following recruitment, participants were fully oriented and provided with a study handbook. Each participant was randomly assigned to consume the four study treatments in a different order from a possible 24 order groups. A total of five participants dropped out of the study because of relocation ($n = 1$), palatability concerns ($n = 2$) and antibiotic use ($n = 2$). A total of 21, 19, 19 and 20 participants completed the potato, chickpea, lentil and green pea treatments, respectively. Participant characteristics at baseline are presented in Table 1.

2.3. Study treatments and diets

The study treatments consisted of three spray-dried powders of pulses (9 mm Kabuli chickpea, large green Laird lentil, and whole green pea), each of 100 g dry weight, and a dehydrated potato-flake control, of 50 g dry weight. The pulses were spray-dried to achieve the aim of having treatments which were uniform, stable throughout the study and easy to rehydrate. The dried pulses were produced at The Food Research and Development Centre of Agriculture and Agri-Food Canada (AAFC) in St. Hyacinthe, QC, in collaboration with Natural Resources Canada. The chickpeas and green peas were soaked in water (1 kg pulse per 1 l water) for 12 h at 4 °C, whereas the lentils were immersed in water and immediately processed. All soak, wash and cooking water was retained and further processed to ensure oligosaccharides retention in each study treatment. The chickpeas, lentils and green peas were cooked for 60, 30–45 and 90–105 min, respectively. Thereafter, they were mixed and homogenized using an Urschel cutter and Microcut Stephan homogenizer before being preheated in a vertical tubular heat exchanger at 35–40 °C. Then, the homogenized wet material was dried in a fast-spouted bed dryer with 4 mm Teflon beads. The differences in processing parameters between the pulses were required to ensure production of uniform homogenized slurries which did not clog the spray dryer. The dried powdered pulses were filtered using 500 and 280 µm sieves and then proportioned in 100 g quantities into individual plastic containers heat-sealed with polyethylene covers. Fifty grams commercial potato flakes were used due to the inability to develop a suitable

Table 1
Participant baseline characteristics^a.

	Mean ± SD
Age (year)	28.1 ± 5.9
Height (cm)	177.7 ± 9.9
Body weight (kg)	79.3 ± 11.6
BMI (kg/m ²)	25.2 ± 3.5
Body fat (%)	18.2 ± 5.3
Waist circumference (cm)	87.1 ± 9.9
Hip circumference (cm)	101.0 ± 6.8
Waist/hip ratio	0.9 ± 0.1
Systolic BP (mm Hg)	125.5 ± 9.1
Diastolic BP (mm Hg)	78.9 ± 7.3
Heart rate (bpm)	69.2 ± 10.4

^a $n = 21$.

spray-dried potato treatment and because 100 g potato flakes produced an unreasonably large volume of product for single-serving consumption upon rehydration.

Participants were instructed to rehydrate the treatment powders as 'soups' but encouraged to experiment with other forms of preparation, including but not limited to, hummus, shakes, or pancakes, with spices and other ingredients, so long as the treatment was consumed in a single serving and preparation details were recorded in the daily study diary. Empty treatment packages were returned at each study visit to assess compliance. Participants were instructed to avoid natural health products, probiotic and prebiotic containing foods, fermented foods and, if possible, over-the-counter medications. Participants were also instructed to limit their consumption of high-fiber foods, such as whole grains and fiber-enriched beverages, to 2–3 times per week, and to limit alcohol intake to less than seven drinks per week and less than two drinks per sitting. Otherwise, participants maintained their pre-study lifestyle, including dietary and physical activity habits, for the duration of the study.

2.4. Data collection

2.4.1. Compositional analysis of study treatments

A standard proximate analysis was performed to quantify the macronutrient content of the pulse and potato study treatments. Moisture was determined in a vacuum oven by drying at 80 °C. Protein was determined by Dumas combustion method, using $N \times 6.25$ to calculate total protein (AOAC method 997.02, 2005). Lipids were extracted using hexane in a Soxtec Avanti (Foss Tecator, Eden Prairie, MN). Ash was determined by charring in a muffle furnace at 525 °C overnight. Glucose was measured spectrophotometrically according to an automated glucose-oxidase method (AOAC method 996.11, 2005). Starch was determined by enzymatically degrading it to glucose with α -amylase and amyloglucosidase and measuring glucose using automated glucose-oxidase method (AOAC method 996.11, 2005).

Dietary fiber was measured gravimetrically (AOAC method 991.43, 2005). This method does not measure oligosaccharides or resistant starch, so these dietary fiber components were measured separately. Sugars and oligosaccharides were extracted from the samples by refluxing in 70% ethanol, following the method used by Bainy, Tosh, Corredig, Poysa, & Woodrow (2008). Concentrations were determined using anion exchange high performance liquid chromatography coupled with pulsed amperometric detection (HPAEC-PAD) with standard curves of glucose, sucrose, raffinose, stachyose and verbascose. To measure resistance starch, samples were incubated with digestive enzymes for 16 h. The remaining starch was hydrolyzed with α -amylase and amyloglucosidase and glucose concentrations were measured spectrophotometrically with an automated glucose-oxidase assay (AOAC method 2002.02, 2005). Table 2 shows the nutrient composition of the study treatments, including the oligosaccharides present.

Table 2
Nutrient composition of treatment soups^a.

	Potato	Chickpea	Lentil	Green pea
Energy (kcal)	149.9	282.7	263.7	245.2
Total carbohydrate (g)	76.8	69.3	66.2	69.2
Dietary fiber (g)	3.5	19.9	14.6	18.4
Resistant starch (g)	0.4	4.4	4.8	5.2
Oligosaccharides ^b (g)	0.0	1.9	2.6	3.5
Protein (g)	2.7	18.3	22.7	21.3
Lipids (g)	0.1	5.0	0.7	0.6
Moisture (g)	3.9	7.0	5.7	6.3
Ash (g)	1.9	2.8	2.6	2.7

^a 100 g for each pulse and 50 g for potato.

^b Includes stachyose, verbascose and raffinose.

2.4.2. Anthropometric measurements

Height was measured on Study Day 1 of the first treatment period using a stadiometer (SECA Portable Stadiometer 214, Hanover, Maryland, USA) and body weight was measured on Study Days 1 and 29 of each treatment period using a digital scale (Acculab SV-100, Edgewood, New York, USA). BMI was calculated as body weight in kg (measured on specific Study Day)/(height in m²) (measured on Study Day 1) for Study Days 1 and 29 of each treatment period. Percent body fat was measured on Study Days 1 and 29 of each treatment period using bioelectrical impedance analysis (BIA) (BodyStat™ 1500, BodyStat Ltd., Detroit, MI). To optimize the accuracy of the BIA, participants were asked to report to Study Day 1 and 29 visits having fasted for 12 h, and HAVING consumed 500 ml of water the evening before, and 500 ml of water the morning before the measurement.

Waist and hip circumferences were measured on Study Days 1 and 29 of each treatment period, with participants standing tall, arms at their sides, using the same spring loaded, inelastic tape measure each time. Waist circumference was measured upon normal exhalation on direct skin and was defined as the centre point between a participant's bottom lower rib and the top of the pelvic bone. Hip circumference was measured over the clothes, with the tape measure pulled tight at the widest point of the hips. Finally, blood pressure (BP) and heart rate (HR) were measured in duplicate on Study Days 1 and 29, of each treatment period, using a digital blood pressure monitor (UA-767PC blood pressure monitor, A&D Medical, Milpitas, CA).

2.4.3. Gastrointestinal questionnaire (GIQ)

GIQs were completed once per day throughout the PRE (Study Days –7 to –1), EARLY (Study Days 1–7) and LATE (Study Days 22–28) phases of each treatment period. Each GIQ consisted of four sections with questions probing: (1) flatulence, (2) abdominal comfort (bloating, cramping, and abdominal discomfort), (3) bowel habits (number of bowel movements, consistency of bowel movements, urgency of bowel movements, problems with defecation), and (4) overall GI function. Definition of terms were provided as follows: flatulence, *the state of having excessive stomach or intestinal gas*; bloating, *a condition in which the abdomen feels full and tight, not only after eating, and often interpreted as excessive intestinal gas*; cramping, *a type of pain that comes and goes or changes in position or severity. It can be very uncomfortable but is rarely serious if it is relieved by passing gas or having a bowel movement*; and abdominal discomfort, *a feeling of discomfort or pain located in the upper or lower abdominal area*.

Participants were asked to indicate the occurrence of a symptom (i.e. 'yes' or 'no') and then, if they answered 'yes', to rank the symptom's severity using a 6-in. horizontal line visual analog scale (VAS) with anchors at 0 and 6 in. The VAS approach has been widely used in the measurement of health outcomes, including pain severity (Akhtar-Danesh, 2001), and recently has been shown effective in the measurement of symptoms of irritable bowel syndrome (Bengtsson, Ohlsson, & Ulander, 2007). The VAS anchors were 'none' and 'worst possible' for questions about abdominal discomfort, bloating, cramping, flatulence and problems with defecation; 'very loose' and 'very hard' for questions about the consistency of bowel movements; 'not urgent' and 'worst possible' for questions about urgency of bowel movements; and 'perfect' and 'worst possible' for questions about overall GI function. Number of bowel movements was determined by asking participants to record the exact number for each day.

2.5. Data and statistical analysis

All statistical analyses were performed using the Statistical Analysis System (SAS, version 9.1, SAS Institute, Cary, NC, USA), with a significance level of $p \leq 0.05$.

The effect of study treatment on participant characteristic data (body weight, BMI, body fat, waist and hip circumferences, BP, HR) was determined using repeated measures ANOVA on Study Day 1 values (controlling for participant and treatment) and repeated measure analysis of covariance (ANCOVA) on Study Day 29 values (controlling for participant, treatment order and treatment, including the variation in Study Day 1 values as a covariate), both followed by the Tukey's test for multiple comparisons.

For the GIQ data, since chi-square analysis indicated no significant differences in their occurrence or ranking of each symptom between Study Days within each of the PRE, EARLY and LATE phases, data within each phase were averaged across Study Days. The effect of phase was determined by comparing occurrence and severity of each symptom between the PRE and EARLY phases and between the PRE and LATE phases within each treatment separately, using chi-square (for occurrence data) or the NPAR1WAY procedure with the Wilcoxon's test (for severity rank and number of bowel movement data, from subjects who responded 'yes' for occurrence). The effect of treatment was determined by comparing occurrence and severity of each symptom between treatments within each phase separately, using chi-square (for occurrence data) and the NPAR1WAY procedure with the Wilcoxon's test (for severity rank and number of bowel movement data, from subjects who responded 'yes' for occurrence).

3. Results and discussion

3.1. Participant characteristics

A comparison of participant characteristic data at Study Days 1 and 29 (adjusted for the variation in Study Day 1 values) of each treatment are presented in Table 3. At Study Day 1, body weight and BMI were significantly different between the potato and lentil

treatments. At Study Day 29, body weight, BMI and waist circumference were significantly different with the green pea and lentil (but not the chickpea), when compared to the potato treatment and hip circumference was significantly different with the green pea (but not the lentil or chickpea) when compared to the potato treatment. Despite these statistical differences, the magnitude of all the changes was minimal and the trends are not suggestive of an overall effect of pulse or potato consumption. Furthermore, there were no significant differences in percent body fat, waist/hip ratio, systolic and diastolic blood pressure or heart rate between any of the study treatments at either Study Days 1 or 29.

3.2. Gastrointestinal questionnaire (GIQ)

Fig. 1 presents the occurrence of flatulence, bloating, cramping, and abdominal discomfort (Fig. 1A–D, respectively), within each treatment at each treatment phase. Table 4 shows the severity of these symptoms and rank results for consistency and urgency of bowel movements, problems with defecation, and overall GI function.

3.2.1. Flatulence

Although flatulence was reported to occur in both the EARLY and LATE phases with consumption of each type of pulse and the potato control, its occurrence was not significantly higher than that reported for the PRE phase (Study Days –1 to –7 before consumption of the treatments) (Fig. 1A). Similarly, flatulence occurrence did not significantly differ between any of the treatments during the PRE, EARLY or LATE phases (Fig. 1A). Flatulence severity was significantly different among treatments during the EARLY ($p < 0.0001$) and LATE ($p < 0.0001$) phases and was ranked significantly higher with consumption of the chickpea and lentil treatments during the EARLY ($p = 0.0004$ and $p = 0.0003$, respectively)

Table 3
Participant characteristics on Study Days 1 and 29 of each treatment^{A,B}.

	Potato (n = 21)	Chickpea (n = 19)	Lentil (n = 19)	Green pea (n = 20)
<i>Body weight (kg)</i>				
Study Day 1	78.9 ± 10.8 ^a	79.8 ± 10.0 ^{ab}	80.8 ± 10.4 ^b	79.5 ± 11.6 ^{ab}
Study Day 29	78.9 ± 10.7 ^a	79.7 ± 10.3 ^a	80.7 ± 10.0 ^b	79.9 ± 11.1 ^{bc}
<i>BMI (kg/m²)</i>				
Study Day 1	25.0 ± 3.2 ^a	25.3 ± 3.23 ^{ab}	25.4 ± 3.5 ^b	25.3 ± 3.7 ^{ab}
Study Day 29	25.0 ± 3.2 ^a	25.2 ± 3.3 ^{ab}	25.4 ± 3.4 ^b	25.4 ± 3.4 ^b
<i>Body fat (%)</i>				
Study Day 1	17.7 ± 4.8	17.9 ± 5.8	17.7 ± 5.0	17.8 ± 4.9
Study Day 29	17.9 ± 5.2	18.1 ± 4.9	18.5 ± 4.9	17.9 ± 5.2
<i>Waist circumference (cm)</i>				
Study Day 1	86.1 ± 8.6	87.4 ± 9.8	87.6 ± 9.2	87.4 ± 10.0
Study Day 29	86.1 ± 8.7 ^a	88.0 ± 8.8 ^{ab}	89.1 ± 8.7 ^b	88.5 ± 9.6 ^b
<i>Hip circumference (cm)</i>				
Study Day 1	100.3 ± 5.1	101.0 ± 6.6	101.6 ± 7.6	100.5 ± 7.3
Study Day 29	99.4 ± 4.8 ^a	101.2 ± 7.4 ^{ab}	101.5 ± 5.9 ^{ab}	101.9 ± 5.7 ^b
<i>Waist/hip ratio</i>				
Study Day 1	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1
Study Day 29	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1
<i>Systolic BP (mm Hg)</i>				
Study Day 1	124.3 ± 9.1	126.4 ± 9.5	126.2 ± 8.9	127.5 ± 13.8
Study Day 29	125.4 ± 10.4	120.6 ± 27.6	128.8 ± 10.5	126.8 ± 11.1
<i>Diastolic BP (mm Hg)</i>				
Study Day 1	78.6 ± 7.2	77.4 ± 8.6	79.0 ± 7.7	78.2 ± 9.5
Study Day 29	78.1 ± 7.4	77.0 ± 8.3	77.7 ± 7.4	78.5 ± 9.7
<i>Heart rate (bpm)</i>				
Study Day 1	69.9 ± 11.1	67.4 ± 14.9	67.7 ± 9.9	70.3 ± 10.0
Study Day 29	67.6 ± 13.9	70.8 ± 13.9	69.0 ± 10.7	68.1 ± 11.2

^A Values are mean ± SD.

^B Values within a row with different superscript letters are significantly different ($p \leq 0.05$).

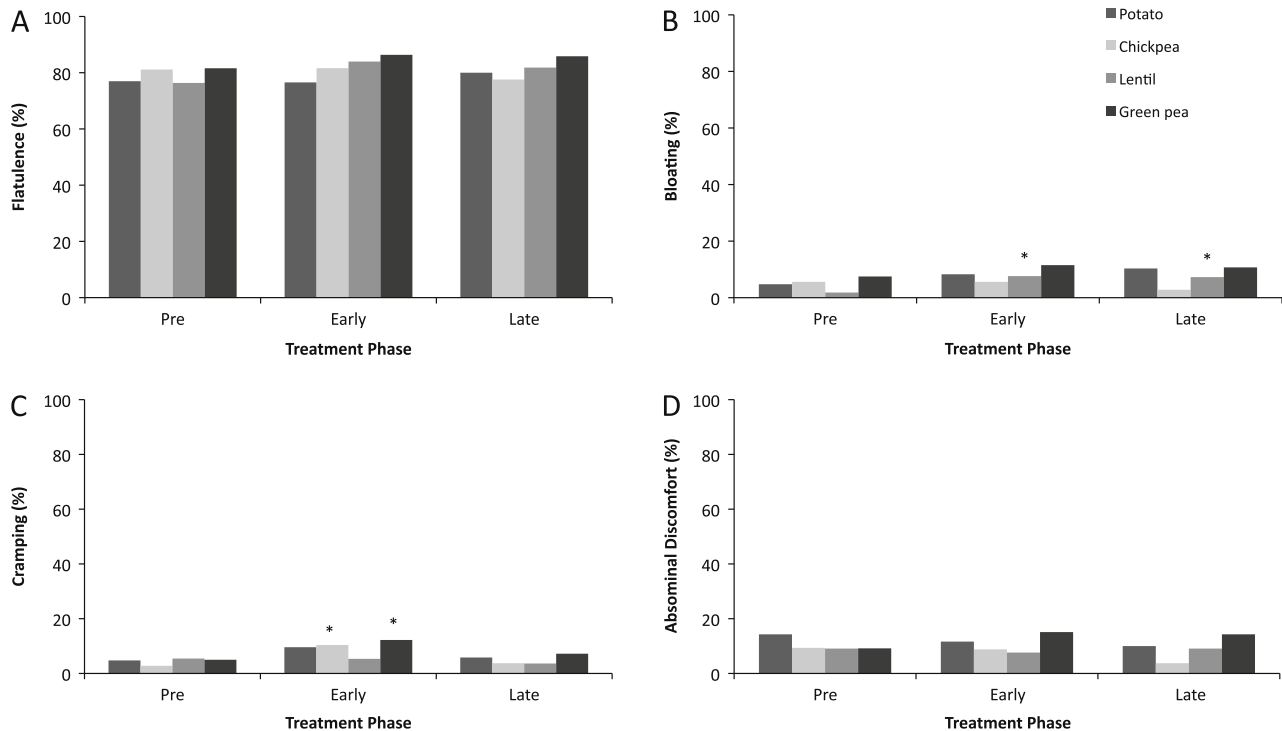


Fig. 1. Percentage of participants reporting experiencing flatulence (A) bloating (B), abdominal cramping (C) and abdominal discomfort (D) within each treatment phase (PRE, EARLY, LATE) for each study treatment (potato, chickpea, lentil, green pea). Values are reported as percent of participants responding 'yes' for symptom occurrence. Significantly different from the PRE phase ($p \leq 0.05$).

and LATE ($p < 0.0001$ and $p = 0.0001$, respectively) phases when compared to the PRE phase (Table 4). These results suggest that flatulence production may depend on pulse type, an idea that is consistent with previous work comparing pulses for their *in vitro* gas (Garg et al., 1980) and *in vivo* flatus (Fleming, 1981) production. However, Table 4 shows that flatulence severity rankings were always very low (i.e. closer to THE 'none' (0 in.) versus 'worst possible' (6 in.) VAS anchor, regardless of treatment or phase.

Despite the small differences observed, pulse consumption was not associated with large changes in participants' perception of the occurrence or severity of flatulence. While increased gas (measured by breath and flatus gas production) has been previously reported with daily consumption of 100 g dry weight red kidney beans for 23 days (O'Donnell & Fleming, 1984), the same study also concluded that, based on subjective responses, the beans were well tolerated by both regular bean consumers and non-bean consumers. Therefore, the current study provides further support to the conclusion that regular pulse consumption does not increase flatulence to a degree that warrants the exclusion of pulses from the diets of healthy adults.

3.2.2. Abdominal comfort

To study the effects of pulse consumption on abdominal comfort, symptoms of bloating, cramping and abdominal discomfort were assessed. There were no significant differences in the occurrence or severity of bloating, cramping or abdominal discomfort between any of the treatments (within each phase) with the exception of a significant effect of treatment on bloating during the EARLY ($p = 0.05$) and LATE ($p = 0.04$) phases. Comparisons between the phases (within each treatment) revealed that bloating occurrence was significantly different during the EARLY ($p = 0.04$) and LATE ($p = 0.05$) phases when compared to the PRE phase during consumption of the lentil treatment (Fig. 1B). Although bloating severity did not significantly differ between the PRE and either the EARLY or LATE phases during consumption of the lentil treat-

ment, it did significantly increase during the EARLY compared to the PRE phase during consumption of the green pea treatment ($p = 0.01$) (Table 4).

Cramping occurrence did not significantly differ between phases for the lentil and potato treatments but did differ between phases for the chickpea and green pea treatments. During consumption of the chickpea treatment, cramping occurrence was significantly higher during the EARLY when compared to the PRE phase ($p = 0.02$) and during consumption of the green pea treatment, cramping occurrence was significantly higher during the EARLY compared to PRE phase ($p = 0.04$) (Fig. 1C). Despite this, there were no significant differences in cramping severity between the PRE and either the EARLY or LATE phases during consumption of any of the treatments (Table 4). Although the chickpea and green pea treatments had the highest dietary fiber contents they did not have the highest levels of oligosaccharides (Table 2), suggesting that a component other than the oligosaccharides may have been responsible for any increases in cramping occurrence or severity.

Abdominal discomfort occurrence (Fig. 1D) and severity (Table 4) were low and did not significantly differ between treatments (within each phase) or phases (within each treatment). Collectively, these results agree with a previous report that fermented beans were well tolerated by healthy adult females with no differences in abdominal pain or bloating rankings (using a nine point hedonic scale) between fermented and non-fermented beans (Granito et al., 2005).

3.2.3. Bowel movements

Overall, none of the pulse treatments affected bowel movements. Number of bowel movements was most commonly reported at a median of 1 per day (data not shown) and did not significantly differ between study treatments (within each phase) or between phases (within each treatment). Similarly, bowel movement consistency, urgency and problems with defecation did not significantly differ between study treatments (within each

Table 4

Gastrointestinal symptom severity scores during the PRE, EARLY and LATE phases of the 28-day treatment periods of potato and pulse (chickpea, lentil, green pea) consumption^{a,b}.

	PRE	EARLY	LATE
<i>Flatulence^c</i>			
Potato	1.19 (0.34, 5.09)	1.13 (0.31, 4.75)	1.44 (0.28, 4.56)
Chickpea	1.22 (0.13, 4.75)	2.11 [*] (0, 5.00)	2.56 [*] (0, 5.69)
Lentil	1.38 (0.25, 4.50)	2.03 [*] (0.22, 4.81)	2.27 [*] (0, 5.00)
Green pea	1.50 (0, 4.34)	1.42 (0, 5.06)	1.78 (0.03, 5.38)
<i>Bloating^c</i>			
Potato	1.42 (0.69, 2.50)	1.50 (0.50, 3.00)	1.55 (1.13, 2.81)
Chickpea	1.02 (0, 3.31)	0.94 (0.47, 1.59)	1.47 (1.44, 1.81)
Lentil	2.63 (1.59, 3.66)	1.94 (0.81, 3.63)	2.77 (1.19, 3.50)
Green pea	1.00 (0.59, 1.69)	1.81 [*] (0.38, 3.84)	1.23 (0, 2.28)
<i>Cramping^c</i>			
Potato	1.33 (0.69, 3.59)	1.53 (0.75, 4.69)	1.63 (0.97, 4.06)
Chickpea	2.16 (0.72, 4.00)	1.34 (0.38, 3.75)	0.78 (0.69, 0.94)
Lentil	1.89 (1.28, 3.16)	1.34 (0.59, 2.22)	1.70 (1.50, 2.81)
Green pea	1.08 (0.72, 1.94)	1.25 (0.63, 3.13)	1.39 (0.66, 2.97)
<i>Abdominal discomfort^c</i>			
Potato	1.34 (0.31, 3.69)	1.25 (0.50, 4.56)	1.72 (0.63, 4.50)
Chickpea	1.50 (0.28, 4.41)	1.53 (0.19, 3.72)	0.77 (0.47, 2.25)
Lentil	1.63 (0.91, 3.06)	1.19 (0.81, 2.88)	1.69 (0.81, 3.31)
Green pea	1.00 (0.50, 2.16)	1.06 (0, 2.81)	1.06 (0.63, 4.97)
<i>Consistency of bowel movements^d</i>			
Potato	2.31 (0.47, 4.63)	2.50 (0, 5.59)	2.66 (0.44, 4.63)
Chickpea	2.39 (0, 5.13)	2.38 (0.19, 5.31)	2.30 (0, 5.03)
Lentil	2.81 (0.28, 5.97)	2.41 [*] (0.28, 5.25)	2.44 [*] (0, 4.53)
Green pea	2.50 (0, 5.41)	2.48 (0, 5.34)	2.30 (0, 5.00)
<i>Urgency of bowel movements^d</i>			
Potato	1.67 (0, 5.00)	1.56 (0, 4.81)	1.69 (0, 5.00)
Chickpea	1.50 (0, 5.00)	1.72 (0, 4.66)	2.09 (0, 4.38)
Lentil	1.48 (0, 4.63)	1.72 (0, 4.78)	1.72 (0, 5.09)
Green pea	1.50 (0, 4.75)	1.64 (0, 4.53)	1.94 (0, 5.25)
<i>Problems with defecation^d</i>			
Potato	0.94 (0, 4.56)	0.94 (0, 6.00)	1.09 (0, 4.31)
Chickpea	0.75 (0, 4.47)	0.75 (0, 4.25)	1.05 (0, 3.84)
Lentil	0.81 (0, 3.75)	0.69 (0, 4.25)	0.94 (0, 4.66)
Green pea	1.13 (0, 4.47)	1.05 (0, 4.94)	1.02 (0, 5.25)
<i>Overall gastrointestinal function^e</i>			
Potato	1.38 (0, 4.47)	1.53 (0, 4.66)	1.19 (0, 4.44)
Chickpea	1.00 (0, 4.63)	1.50 (0, 4.69)	1.50 (0, 6.00)
Lentil	1.11 (0, 4.00)	1.13 (0, 4.47)	1.19 (0, 4.91)
Green pea	1.20 (0, 4.66)	1.23 (0, 4.81)	1.23 (0, 5.31)

^a Significantly different from the PRE phase ($p \leq 0.05$).

^b PRE refers to Study Days –7 to –1; EARLY refers to Study Days 1–7; LATE refers to Study Days 22–28.

^c Values are median scores in inches measured to the nearest one eighth inch (minimum, maximum).

^d Comparisons were performed only with data from participants who responded 'yes' to occurrence of the symptom.

^e Comparisons were performed only with data from participants who reported >0 bowel movements.

^f Comparisons were performed on data from all participants.

phase) or between phases (within each treatment) except that bowel movement consistency was significantly lower in the EARLY ($p = 0.03$) and LATE ($p = 0.01$) phases compared to the PRE phase during consumption of the lentil treatment (Table 4). Pulses are high in fiber, an essential part of the diet believed to promote laxation (increased frequency of bowel movements and softening of the stool) and normalize defecation frequency to one bowel movement per day (Slavin, 2008). The subjective nature of the bowel movement consistency measurement in the current study could have made it difficult to detect subtle bowel movement softening. However, importantly, there were no reports of extremely hard or liquid bowel movements, both indicators of abnormal bowel function (Slavin, 2008). Also, prior to beginning the study, participants reported, on average, regular bowel movements, which may explain why minimal significant changes in number of bowel move-

ments were observed. Consistent with these results, a previous study found that 23 days of 100 g dry weight kidney bean consumption did not significantly affect habitual fecal output or transit time (Fleming, O'Donnell, & Perman, 1985).

3.2.4. Overall gastrointestinal function

Rankings of overall GI function did not significantly differ between treatments (within each phase) or phases (within each treatment) (Table 4). These findings lend support to the argument that some level of dietary LDC, such as the GOS found in pulses, can be tolerated. The results are in agreement with reports that consumption of 45 g cooked dried white beans were widely accepted (76.5%) by a sample of 51 women (Granito et al., 2005). There is debate over the upper tolerable limit for LDC (Bouhnik et al., 1999; Marteau & Seksik, 2004; Martinez-Villaluenga et al., 2008). For example, it has been suggested that adverse GI effects are only experienced when LDC are consumed at high doses (i.e. more than 20 g per day) (Marteau & Seksik, 2004). Conversely, it was shown that 10 g short chain fructo-oligosaccharides per day, as part of a regular diet, were well tolerated and led to an increase in colonic bifidobacteria (Bouhnik et al., 1999). More recently, it has been proposed that as little of 3 g of alpha-oligosaccharides per day may offer health benefits associated with a prebiotic effect without causing GI distress (Martinez-Villaluenga et al., 2008). The present study delivered 1.9, 2.6 and 3.5 g pulse GOS per day for the chickpea, lentil and green pea treatments, respectively and up to 19.9 g dietary fiber (Table 2), without inducing major changes in GI symptoms. However, continued research is necessary since the level of tolerance to LDC depends on many factors (i.e. the amount in food consumed, relative concentration of the LDC, frequency of consumption, as well as bowel habit, including extent of bowel fill and transit time (Livesey, 2001)), which may account for variability between individuals and studies.

4. Conclusion

The present study found that 28 consecutive days of consumption of spray-dried Kabuli chickpeas, green Laird lentils or green peas, containing oligosaccharides, was well tolerated by a sample of 21 healthy adult males. The modest effects observed on severity of flatulence and GI symptoms resulting from the quantity of pulses consumed are not likely to cause sufficient concerns as to warrant limiting or avoiding pulse consumption. Therefore, these findings help to dispel the myth that intolerable levels of flatulence and GI discomfort occur as a result of pulse consumption. Future research should continue to examine the link between pulse composition and GI effects, including whether or not specific pulse varieties are more suitable for incorporation into novel functional foods.

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